

Error Analysis for Iteration Methods

① Order of Convergence

$$\text{Formula} = \boxed{\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = L}$$

linearly $\rightarrow \alpha = 1$

quadratically $\rightarrow \alpha = 2$

cubically $\rightarrow \alpha = 3$

When (α) is bigger number that's mean your approach to $(p) \rightarrow$ Root

Q6 : 6. Show that the following sequences converge linearly to $p = 0$. How large must n be before $|p_n - p| \leq 5 \times 10^{-2}$?

a. $p_n = \frac{1}{n}, \quad n \geq 1$

$$= \lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = L$$

$\Rightarrow$ If we have p_n we can find p_{n+1} : $p_n = \frac{1}{n} \rightarrow p_{n+1} = \frac{1}{n+1}$

from given in Q $\Rightarrow p = 0$, $\alpha = \text{linearly} = 1$

to solve limits ignore (L)

$$= \lim_{n \rightarrow \infty} \frac{|\frac{1}{n+1} - 0|}{|\frac{1}{n} - 0|^1}$$

$$= \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{n+1} \right|}{\left| \frac{1}{n} \right|} \Rightarrow \lim_{n \rightarrow \infty} \frac{\frac{1}{n+1}}{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n+1} \cdot n$$

$$= \lim_{n \rightarrow \infty} \frac{n}{n+1} \quad \text{"to solve limit use L'Hopital rule"}$$

by dividing the highest degree of (n)

$$= \lim_{n \rightarrow \infty} \frac{\frac{n}{n}}{\frac{n}{n} + \frac{1}{n}}$$

$$\text{Remark } \circ \rightarrow \lim_{n \rightarrow \infty} \frac{n}{n} = 1$$

$$\rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$$

$$= \lim_{n \rightarrow \infty} \frac{1}{1+0} = \boxed{1}$$

So P_n is Converges linearly in $p=0$

$$\text{b. } p_n = \frac{1}{n^2}, \quad n \geq 1$$

$$\Rightarrow \text{If we have } p_n \text{ we can find } p_{n+1} \circ \quad p_n = \frac{1}{n^2} \Rightarrow p_{n+1} = \frac{1}{n^2+1}$$

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{1}{n^2+1} - 0 \right|}{\left| \frac{1}{n^2} - 0 \right|} \Rightarrow \lim_{n \rightarrow \infty} \frac{\left| \frac{1}{n^2+1} \right|}{\left| \frac{1}{n^2} \right|} \Rightarrow \lim_{n \rightarrow \infty} \frac{\frac{1}{n^2+1}}{\frac{1}{n^2}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^2+1} \cdot \frac{n^2}{1} \Rightarrow \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} \quad \text{"use L'Hopital rule"}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n^2}{n^2}}{\frac{n^2}{n^2} + \frac{1}{n^2}} \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{1+0} = \boxed{1}$$

So P_n is Converges linearly in $p=0$

Q8: 8. a. Show that the sequence $p_n = 10^{-2^n}$ converges quadratically to 0.
 $\alpha = 2$ $P = 0$

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - P|}{|p_n - P|^\alpha} = L$$

$\Rightarrow$ If we have p_n we can find p_{n+1} : $p_n = 10^{-2^n} \Rightarrow p_{n+1} = 10^{-2^{n+1}}$
 $p_{n+1} = 10^{-2^{n+1}} = 10^{-2^n} \times 10^{-2^n}$

given in Q $\Rightarrow \alpha = 2$ and $P = 0$

$$\lim_{n \rightarrow \infty} \frac{|10^{-2^{n+1}} - 0|}{|10^{-2^n} - 0|^2}$$

$$= \lim_{n \rightarrow \infty} \frac{|10^{-2^{n+1}}|}{|10^{-2^n}|^2}$$

$$= \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{|10^{-2^n}|^2} \Rightarrow \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{10^{-2^n \cdot 2}} \quad \text{"If they've same base we can add powers } 2^n \cdot 2^1 = 2^{n+1}"$$

$$= \lim_{n \rightarrow \infty} \frac{10^{-2^{n+1}}}{10^{-2^{n+1}}}$$

$$\lim_{n \rightarrow \infty} 1 = 1$$

So p_n is Convergent in $P=0$

Q9: 9. a. Construct a sequence that converges to 0 of order 3.

$$p_n = 7^{-3^n}$$

Base can be any number

Should match order α w/ power of p_n

$\Rightarrow$ If we have P_n we can find P_{n+1} : $P_n = 7^{-3^n} \Rightarrow P_{n+1} = 7^{-3^{n+1}}$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{|7^{-3^{n+1}} - 0|}{|7^{-3^n} - 0|^3} \Rightarrow \lim_{n \rightarrow \infty} \frac{|7^{-3^{n+1}}|}{|7^{-3^n}|^3} \Rightarrow \lim_{n \rightarrow \infty} \frac{7^{-3^{n+1}}}{7^{-3^n \cdot 3}}$$

$$\lim_{n \rightarrow \infty} \frac{7^{-3^{n+1}}}{7^{-3^{n+1}}} = 1$$

$$\lim_{n \rightarrow \infty} 1 = 1$$

So P_n is Convergent in $P=0$

Q38

3. Repeat Exercise 1 using the modified Newton's method described in Eq. (2.13). Is there an improvement in speed or accuracy over Exercise 1?

a. $x^2 - 2xe^{-x} + e^{-2x} = 0$, for $0 \leq x \leq 1$

$$\text{modified Newton's method} \Rightarrow P_n = P_{n-1} - \frac{f(P_{n-1}) f'(P_{n-1})}{f(P_{n-1})^2 - f(P_{n-1}) f''(P_{n-1})}$$

$$\Rightarrow f(x) = x^2 - 2xe^{-x} + e^{-2x}, [0, 1]$$